Solutions for Analysis I by Terence Tao

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Contents

Remark. This document follows the third edition of *Analysis I*.

1 Introduction

No exercises from this chapter.

2 Starting at the beginning: the natural numbers

2.1 The Peano axioms

No exercises from this subchapter.

2.2 Addition

Exercise 2.2.1. *Prove Proposition 2.2.5 (Addition is associative): For any natural numbers a, b, c, we have $(a+b)+c = a+(b+c)$.*

Solution. We induct on a, keeping b and c fixed.

First, consider the base case $a = 0$. Then we have $(0+b)+c = 0+(b+c)$. By Lemma 2.2.2, we obtain $b+c = b+c$ as desired.

Now we assume inductively that $(a+b)+c = a+(b+c)$; we now have to show $((a{+}{+})+b)+c = (a{+}{+})+(b+c)$ to close the induction. By applying Lemma 2.2.3 twice on the left-hand side, $((a{+}{+})+b)+c = ((a+b){+}{+})+c = (a+b+c){+}{+}$. Similarly, applying Lemma 2.2.3 on the right-hand side yields $(a{+}{+})+(b+c) = (a+b+c{+}{+}$. Thus, both sides are equal, and we have closed the induction. □

Exercise 2.2.2. *Prove Lemme 2.2.10: Let a be a positive number. Then there exists exactly one natural number b such that $b{+}{+} = a$.*

Solution. We induct on a.

First, consider the base case $a = 1$. By Definition 2.1.3, $1 = 0{+}{+}$. Suppose there is another natural number b such that $b{+}{+} = 1$. However, by Axiom 2.4, if $b{+}{+} = 0{+}{+} = 1$, then $b = 0$. Thus, there is exactly one natural number b such that $b{+}{+} = 1$. Hence, the base case is proven.

Suppose inductively that there exists exactly one natural number b such that $b{+}{+} = a$. We now have to show there exists exactly one natural number b' such that

$b'{+}{+} = a{+}{+}$. By the inductive hypothesis, $a{+}{+} = (b{+}{+}){+}{+} = b'{+}{+}$. As per Axiom 2.4, there exists exactly one solution for b', namely $b' = a$. This closes the induction. $\square$

Exercise 2.2.3. *Prove Proposition 2.2.12 (Basic properties of order of natural numbers).*

Solution. Let a, b, c be natural numbers. Then

(a) *Reflexivity: $a \geq a$.*

By Lemma 2.2.2, $a + 0 = a$. Thus, by Definition 2.2.11, $a \geq a$.

(b) *Transitivity: If $a \geq b$ and $b \geq c$, then $a \geq c$.*

By Definition 2.2.11, write $a = b + m$, $b = c + n$, where m and n are natural numbers. Thus, $a = (c+n)+m = c+(n+m)$ by associativity of addition. By mathematical induction, it is trivial to prove that $n + m$ is a natural number. Therefore, $a \geq c$. $\square$

(c) *Anti-symmetry: If $a \geq b$ and $b \geq a$, then $a = b$.*

By Definition 2.2.11, if $a \geq b$, then for a natural number m, $a = b + m$. But also if $b \geq a$, then $b = a + n$, where n is a natural number. Thus $a = (a+n)+m = a+(n+m)$. By Lemma 2.2.2 and Proposition 2.2.6 (cancellation law), $0 = n + m$. By Corollary 2.2.9, this results in $m = n = 0$. Thus $a = b + 0 = b$ as required. $\square$

(d) *Preservation of order: $a \geq b$ if and only if $a + c \geq b + c$.*

First, prove $a \geq b \implies a + c \geq b + c$. According to Definition 2.2.11, if $a \geq b$, then $a = b + m$, where m is a natural number. Thus $a + c = (b + m) + c = (b + c) + m$, thus $a + c \geq b + c$.

Conversely, consider $a + c = (b + c) + m$, where m is a natural number. By Proposition 2.2.6 (cancellation law) and the associativity of addition, $a = b+m$, meaning $a \geq b$.

Because $a \geq b \implies a + c \geq b + c$ and $a + c \geq b + c \implies a \geq b$, $a + c \geq b + c \iff a \geq b$. $\square$

(e) *$a < b$ if and only if $a{+}{+} \leq b$.*

First, consider $a < b \implies a{+}{+} \leq b$. According to Definition 2.2.11, if $a < b$, then $b = a + m$, where m is a natural number. Note that $m \neq 0$

because we would otherwise obtain $a = b$, which is contradictory to the given condition $a < b$. Therefore, by Axiom 2.3, we have $m = n{+}{+}$, where n is a natural number. Thus, by the definition and commutativity of addition, $b = a + m{+}{+} = (a+m){+}{+} = (a{+}{+}) + m$. It follows that $a \leq b$.

Now consider the converse. From $a{+}{+} \leq b$, we obtain $b = a{+}{+} + p$, where p is a natural number. Thus, by the definition and commutativity of addition, $b = a{+}{+} + p = (a+p){+}{+} = a + p{+}{+}$. According to Axiom 2.3, 0 is not the successor of any natural number, so $p{+}{+} \neq 0$. Hence, $a \leq b$.

Because $a < b \implies a{+}{+} \leq b$ and $a{+}{+} \leq b \implies a < b$, $a < b \iff a{+}{+} \leq b$. $\square$

(f) *$a < b$ if and only if $b = a + d$ for some positive number d.*

First consider $a < b \implies b = a+d$. From $a < b$ we know $a \geq b$, or $b = a+d$ for a natural number d. For the sake of contradiction, suppose $d = 0$. Therefore $a = b$, which contradicts $a < b$. Hence, d is positive; thus, $a < b \implies b = a+d$ for some positive number d.

Now consider the converse. From Definition 2.2.11, $b = a + d$ implies $b \geq a$, or $a \leq b$. If $b = a$, by Proposition 2.2.6 (cancellation law), $d = 0$, which contradicts the given condition that d is positive. Therefore, $a \neq b$, which implies $a < b$.

Because the statement and the converse are both true, we can conclude that for some positive number d, $a < b$ if and only if $b = a + d$. $\square$

Exercise 2.2.4. *Justify the three lemmas marked with (why?) in Proposition 2.2.13 (trichotomy of order for natural numbers).*

Solution.

(a) *For all b, $0 \leq b$.*

From the definition of addition, $0 + b = b$. By Definition 2.2.11, this implies $0 \leq b$. $\square$

(b) *If $a > b$, then $a{+}{+} > b$.*

From Definition 2.2.11, $a = b + m$, where m is a natural number. Thus, by the definition of addition, $a{+}{+} = (b+m){+}{+} = b + m{+}{+}$. Because $m{+}{+}$ is positive by Proposition 2.2.8, $a{+}{+} > b$. $\square$

(c) *If $a = b$, then $a{+}{+} > b$.*

From Axiom 2.4, if $a = b$, then $a{+}{+} = b{+}{+}$. Recall that $b{+}{+} = b + 1$, thus $a{+}{+} = b + 1$. By Definition 2.2.11, this implies $a{+}{+} > b$. □

Exercise 2.2.5. *Prove Proposition 2.14 (Strong principle of induction): Let m_0 be a natural number, and let $P(m)$ be a property pertaining to an arbitrary natural number m. Suppose that for each $m \geq m_0$, we have the following implication: if $P(m')$ is true for all natural numbers $m_0 \leq m' < m$, then $P(m)$ is also true. Then we can conclude that $P(m)$ is true for all natural numbers $m \geq m_0$.*

Solution. Let $Q(n)$ be the property such that $P(m)$ is true for all $m_0 \leq m < n$. Note that the supposed implication becomes "If $Q(n)$ is true, then $P(n)$ is also true. We induct on n.

First, consider the base case $n \leq m_0$. In this case, $Q(n)$ is vacuously true as there is no such natural number m. Hence, the base case is proven.

Now, we assume inductively that $Q(n)$ is true; we will now prove $Q(n{+}{+})$ is true. By the inductive hypothesis, $P(m)$ is true for all m where $m_0 \leq m < n$. According to the supposed implication, because $Q(n)$ is true, $P(n)$ is also true. Therefore, $P(m)$ is true for $m_0 \leq m \leq n$, which implies $Q(n{+}{+})$ is also true. This closes the induction. □

Exercise 2.2.6. *Let n be a natural number, and let $P(m)$ be a property pertaining to the natural numbers such that whenever $P(m{+}{+})$ is true, then $P(m)$ is true. Suppose that $P(n)$ is also true. Prove that $P(m)$ is true for all natural numbers $m \leq n$; this is known as the* principle of backwards induction*.*

Solution. Write $Q(n)$ as the property "If $P(n)$ is true, $P(m)$ is true for all natural numbers $m \leq n$. We apply induction on n.

Consider the base case $n = 0$. The statement $m \leq n$ implies $0 = m + x$ for a natural number x. According to Corollary 2.2.9, this leads to $m = x = 0$. Thus, $Q(n)$ is equivalent to "If $P(n)$ is true, $P(n)$ is true.", which is always true regardless of whether $P(n)$ is true or false. Hence, the base case is proven.

Suppose inductively that $Q(n)$ is true; we now need to show $Q(n{+}{+})$ is true. From the definition of $P(m)$, $P(n{+}{+})$ being true implies $P(n)$ is true. By the inductive statement, $P(m)$ is true for all natural numbers $m \leq n$. Thus if $P(n{++})$ is true, $P(m)$ is true for all $m \leq n{++}$. This is precisely the definition of $Q(n{++})$, so $Q(n{++})$ is also true. This closes the induction. $\square$